Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

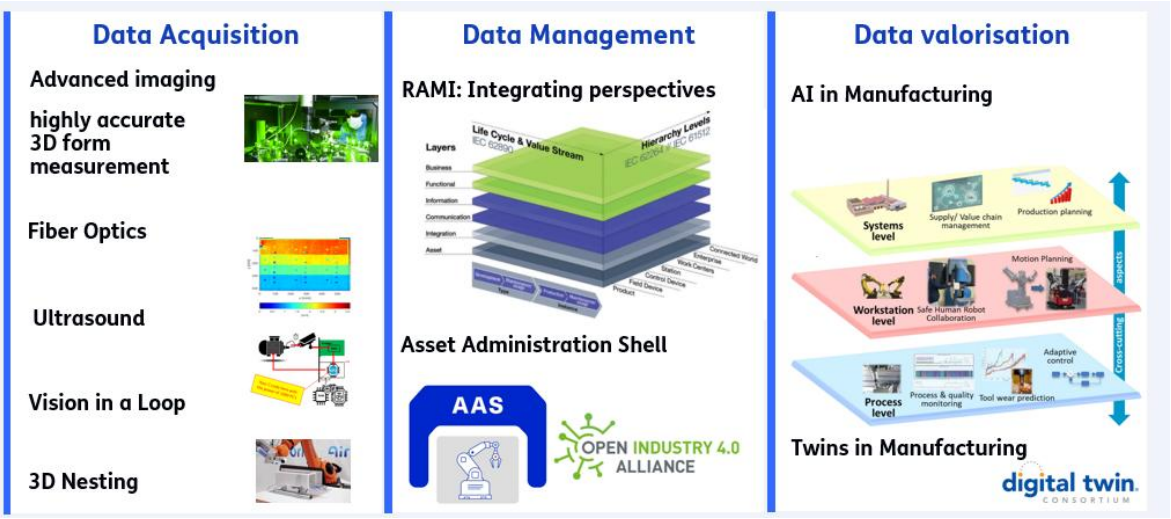
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Mark Courage	Email	mark.courage@tno.nl
Institute Name	TNO Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek TNO (Netherlands Organisation for applied scientific research TNO)			
Address	Anna van Buerenplein 1, 2595 DA The Hague, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Interoperable Digital Twins based Asset Administration Shell (AAS) for Smart Manufacturing and High-Tech Supply Chains  			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Measurement/Sensor Technologies for Automation*	*U-computing*	*U-Computing Platforms and Application Technologies*

Concept of Proposed Technology

Manufacturing companies increasingly rely on high-fidelity digital representations of machines, components, and production systems to shorten engineering cycles, increase uptime, and collaborate efficiently across supply chains. TNO proposes a modular, interoperable approach to Digital Twins (DTs) combined with Asset Administration Shells (AAS) as standardized containers for machine and process information. The technology includes:

Lifecycle-Based Digital Twin Models

- Simulation, monitoring, and prediction capabilities using real-time sensor data.
- Hybrid modelling (physics-based + data-driven), as described in TNO's research on digital twin model architectures.

AAS-Enabled Interoperability

- Standardized machine and component data models enabling plug-and-play integration across factories and IT platforms.
- Connects design, operations, service, and supply chain information.

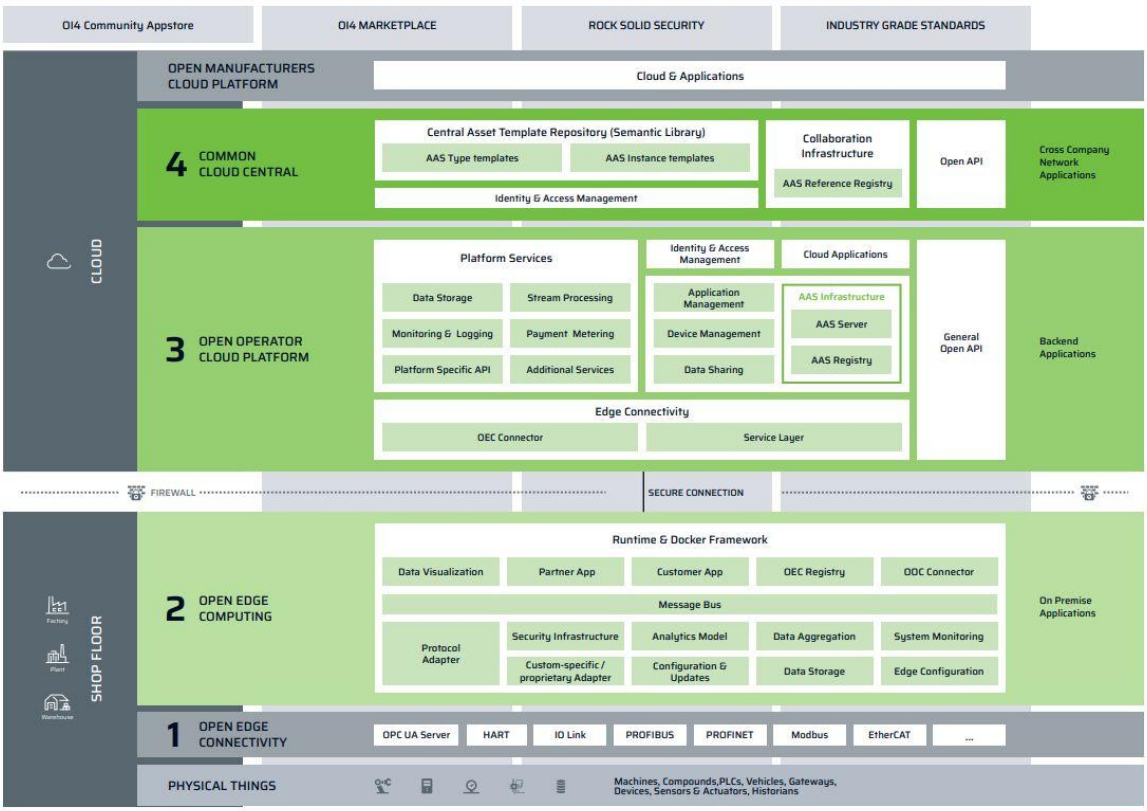
Reference Architectures & Tooling

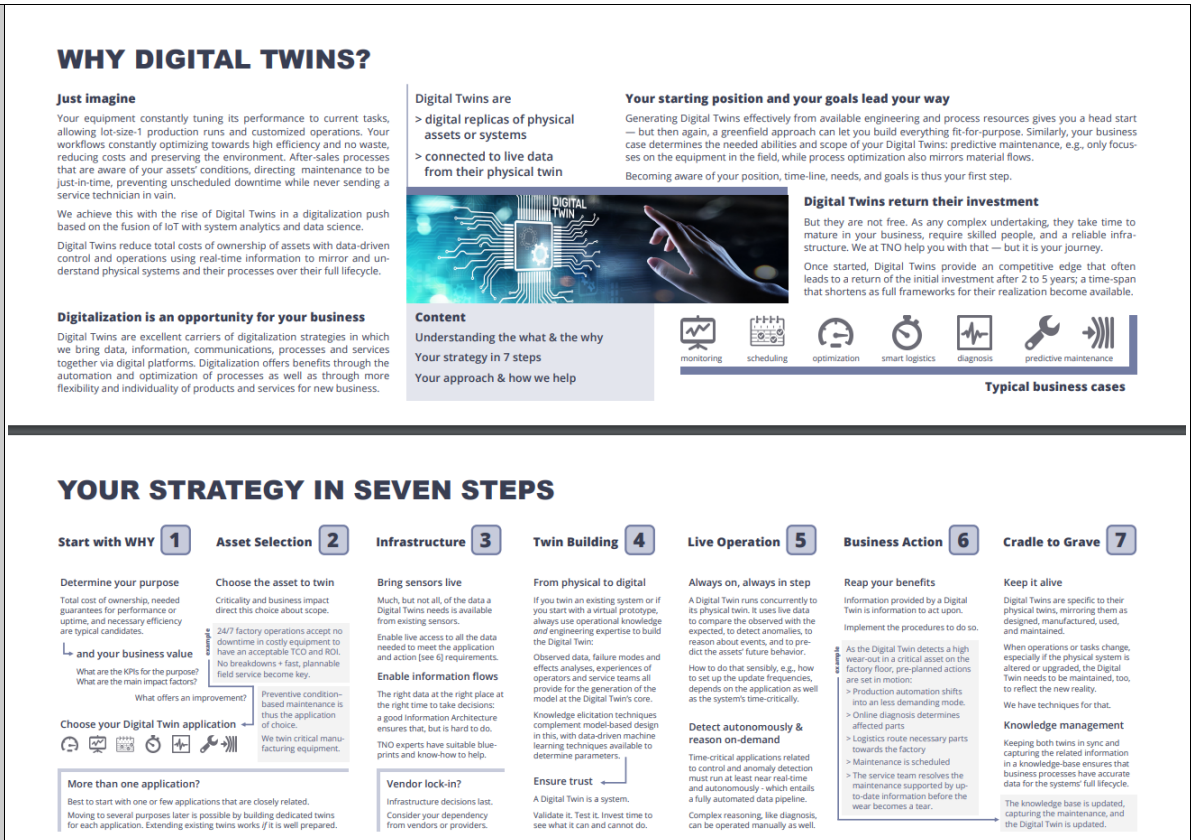
- Model libraries, validation methods, and integration workflows that enable SMEs to develop DTs at lower cost and risk.
- Based on practices developed with the Dutch high-tech manufacturing ecosystem (Brainport Industries).

The combined DT+AAS+RAMI approach allows SMEs to implement actionable, maintainable, and scalable digital solutions supporting first-time-right engineering, predictive maintenance, and cross-company data exchange.

DS.AR AI Model Repository	DS.AG Data Aggregation	IR.AS API Srvices				UX.GM Gamification	UX.DB Dashboards
DS.SR Simulation Model Repository	DS.AS Asynchronous Integration	IR.CL Collaboration Platform Integration	IC.CS Composition	IC.SM Simulation	IC.RP Reporting	UX.3R 3D Rendering	UX.XR Extended Reality (XR)
DS.SA Data Storage and Archive Services	DS.RT Real-time Processing	IR.DT Digital Twin Integration	IC.DL Distributed Ledger and Smart Contracts	IC.FL Federated Learning	IC.AL Alerts and Notifications	UX.GE Gaming Engine Visualization	UX.ER Entity Relationship Visualization
DS.DS Domain Specific Data Management	DS.BP Batch Processing	IR.IO OT/IoT System Integration	IC.BR Business Rules	IC.AI Artificial Intelligence	IC.OS Orchestration	UX.BP Business Process Mgmt & Workflow	UX.RM Real-time Monitoring
DS.IR Digital Twin Instance Repository	DS.CX Data Contextualization	IR.EG Engineering Systems Integration	IC.PS Prescriptive Recommendations	IC.PR Prediction	IC.IC Command and Control	UX.BI Business Intelligence	UX.AV Advanced Visualization
DS.RP Digital Twin Model Repository	DS.TR Data Transformation and Wrangling	IR.ET Enterprise System Integration	IC.MA Mathematical Analytics	IC.AA Data Analysis and Analytics	IC.SR Search	UX.CI Continuous Intelligence	UX.BV Basic Visualization
DS.ON Ontology Management	DS.ST Data Streaming	MG.DG Data Governance	MG.SM System Monitoring	TW.RP Responsibility	TW.RL Reliability	TW.PR Privacy	TW.DS Device Security
DS.SG Synthetic Data Generation	DS.AI Data Acquisition and Ingestion	MG.EL Event Logging	MG.DM Device Management	TW.RS Resilience	TW.SF Safety	TW.SC Security	TW.EX Data Encryption

Data Services · Integration · Intelligence · UX · Management · Trustworthiness

Rationale for Joint R&D Project	 TNO–Korean SME project is essential because: • Korean SMEs in semiconductors, robotics, mobility, and high-precision manufacturing face increasing pressure to deliver traceable, data-rich, digitally compatible equipment to global OEMs. • Digital Twins are complex and resource-intensive, especially regarding model validation, lifecycle management, and interoperability standards. • TNO brings decades of expertise in system engineering, model-based development, AAS implementation, and DT validation methods; Korean SMEs contribute strong engineering capabilities, domain knowledge, and rapid industrial adoption. • Joint development accelerates cross-continental standardization and ensures solutions meet both EU and Asian market requirements. • The project creates shared demonstrators for commercialization in semiconductor equipment, high-tech machinery, robotics, and smart factories.



Key Words	1. Digital Twin (DT) 2. Asset Administration Shell (AAS) 3. Interoperability / Industry 4.0 4. Predictive Maintenance 5. High-Tech Manufacturing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	*TRL 4 (Lab validation of integrated modules)*	**End of the Project**	*TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)*
R&D Objectives	**Key R&D objectives:** • Modular Digital Twin components for equipment and production lines. • AAS-compliant information models for machine and subsystem data exchange. • Integrated DT+AAS demonstrators for predictive maintenance, condition monitoring, and engineering optimization. • Validation and verification (V&V) workflows for industrial deployment. • Reference architecture applicable to semiconductor, robotics, and precision manufacturing SMEs. The overarching R&D objective of this joint TNO–Korean SME initiative is to develop a scalable, interoperable, and industry-validated digital-twin framework based on the Asset Administration Shell (AAS) and internationally recognized Industry 4.0 reference architectures (RAMI 4.0). The project			

	aims to demonstrate how Korean SMEs in sectors such as semiconductor tooling, robotics, and precision equipment manufacturing can adopt and deploy digital twins as part of their core operational and engineering workflows. Central to this objective is the creation of modular digital-twin building blocks, spanning data acquisition, hybrid modelling, semantic data structures, and lifecycle management, that can be reused across multiple SME contexts while enabling customization for specific machine platforms, manufacturing processes, or subsystem architectures. A key component of the R&D objective is that the detailed functional requirements and performance targets will be refined iteratively, based on the concrete use cases defined by participating Korean SMEs. These use cases will be treated as anchor points for the engineering effort, ensuring that the resulting digital-twin framework addresses real operational bottlenecks such as predictive maintenance needs, equipment-to-equipment interoperability, process optimization, and cross-factory data traceability. This use-case-driven approach ensures that the research maintains direct industrial relevance and that the resulting technologies can be validated under real production conditions. The R&D will further integrate principles from system-engineering methodologies, ensuring a structured decomposition of SME capability needs into requirements for sensing, data acquisition, semantic data modelling, simulation, analytics, and interface design. By combining the Open Industry 4.0 Alliance vision, particularly its AAS submodel and interoperability guidelines, with the RAMI 4.0 reference model, the project aims to deliver a harmonized DT–AAS architecture that is compatible with emerging global standards for smart manufacturing. In addition, insights from the Autonomous Systems Maturity Framework (capturing the interplay between technology, ecosystem configuration, and business-model maturity) will be used to determine which level of autonomous behaviour and decision-making is feasible within SME environments, and how digital twins can progressively evolve from operator-assist functionality toward semi-autonomous and eventually self-optimizing manufacturing modules. Ultimately, the R&D objective is to establish a validated, referenceable, and exportable digital-twin architecture that Korean SMEs and TNO can jointly commercialize, helping SMEs to increase data transparency, improve equipment uptime, and meet the growing digital-traceability demands of global OEM customers.
R&D Target Specifications	**Key R&D Target Specifications:** • Digital Twin data ingestion latency: < 1 second (near real-time) • Predictive maintenance model accuracy: > 80% fault prediction precision • AAS-compliant data model coverage: ≥ 90% of key machine parameters • Interoperability: successful connection to at least 3 external systems (MES, ERP, or machine controllers) • Demonstrator system achieving 10–20% reduction in downtime in pilot use cases The R&D target specification defines the intended performance, interoperability, and functional capabilities of the DT–AAS integrated solution developed in this project. These targets will be progressively elaborated and optimized based on the specific characteristics and needs of the Korean SME use cases selected during the project's initial phase. This adaptive, use-case-anchored specification process ensures that the framework is not only theoretically robust but also practically implementable in diverse industrial contexts. At the highest level, the R&D targets include the development of a **generic but configurable digital-twin framework built on AAS submodels** compliant with Open Industry 4.0 and aligned to the RAMI 4.0 layers (Asset, Integration, Communication, Information, Functional, and Business).

	The framework should enable Korean SMEs to achieve seamless interoperability across heterogeneous machine fleets, subsystems, and digital platforms, reducing the integration burden and enabling plug-and-play data exchange. This includes well-defined targets for semantic data quality, real-time data ingestion latency (<1 second for selected use cases), standardized AAS submodel coverage (>90% of asset-relevant parameters), and validated connectors to external systems such as MES, ERP, SCADA, and equipment controllers. In accordance with a system-engineering approach, the target specification encompasses functional and non-functional requirements for each layer of the digital-twin architecture: • **Data Acquisition Targets:** Define sensing, networking, and protocol requirements based on SME equipment constraints; ensure robust data capture from high-precision sensors, industrial IoT devices, and existing machine controllers. • **Semantic Interoperability Targets:** Implement harmonized AAS submodels for predictive maintenance, condition monitoring, machine configuration, and lifecycle management, following Open Industry 4.0 guidelines. • **Model-based Engineering Targets:** Achieve hybrid physics-informed + data-driven model accuracy for fault prediction (>80% fault-prediction precision where applicable) and process optimization. • **Operational Performance Targets:** Demonstrate measurable improvements (10–20% reduction in downtime, 5–15% reduction in engineering-change time, or KPI improvements tailored to specific SME use cases). • **Ecosystem Integration Targets:** Establish the capability to connect at least three external systems and support automated data exchange between multiple SMEs participating in a shared pilot or reference use case.' Additionally, the specification incorporates insights from the Autonomous Systems Maturity Framework to determine feasible levels of automation and decision support for SME environments. The project will define incremental digital-twin capability levels—ranging from operator assistance to semi-autonomous feedback loops—aligned with the SMEs' process complexity, data maturity, and operational safety requirements. This ensures that the target specification supports both immediate value delivery and long-term evolution toward higher maturity levels.
R&D Plan	**Phase 1 – Requirements, Architecture & Modelling (Months 1–4)** • Select use cases (e.g., semiconductor tool subsystem, robotics module). • Define reference architecture combining DT + AAS. • Create AAS submodels for assets, components, and processes. **Phase 2 – Model Development & AAS Integration (Months 5–12)** • Build hybrid digital twin models (physics & ML). • Implement real-time connectors to equipment and sensors. • Integrate AAS service layers for standardized communication. **Phase 3 – Validation, Pilot Integration & Demonstrator (Months 12–24)**

	• Perform laboratory validation with test equipment. • Deploy pilot in SME environment (Korea or NL). • Measure performance: prediction accuracy, interoperability, engineering time reduction. **Phase 4 – Finalization & Commercialization Package (Months >24)** • Deliver reference implementation, documentation, and training. • Prepare joint commercialization plan for EU–Asia markets.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide DT architecture, modelling methods, and AAS implementation expertise. • Develop model libraries, validation tools, and reference frameworks. • Lead system integration, testing, and verification. • Deliver standardization guidance (AAS, OPC UA, interoperability profiles). • Train SME partners in model-based engineering workflows.
Expected R&R of Korean SME Partner	• Provide domain-specific machine or manufacturing use cases. • Share data, operational insights, and equipment access for model development. • Implement prototypes in real environments and perform testing. • Contribute to productization, UI/UX, and customer requirements. • Participate in co-commercialization in Korea and Asia.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Development of interoperable, scalable Digital Twin solutions ready for commercialization. • AAS-based data structures enabling cross-company collaboration and reducing integration effort by 30–50%. • Increased equipment uptime and lower maintenance costs (10–20% reduction). • Strengthening of Korea–Netherlands cooperation in high-tech manufacturing and smart factory technology. • Creation of exportable joint offerings for semiconductor, robotics, and advanced manufacturing markets. • Empowerment of Korean SMEs to meet future OEM requirements for digital traceability and lifecycle data integration.

6. References

(1) *TNO Digital Twinning:* Digital Twins in practice | TNO